

Q1. Calculation of Margin

$$\frac{12 - c}{12} = 0.45 \quad 12 - c = 5.4$$

Margin is £5.4 per product

Avg. consumer purchases 3 a year So average Margin of £16.2 per customer

$$M = 16.2 \quad i = 0.7$$

Calculation of retention rate

$$L = \frac{1}{1-r} \quad 2.33 = \frac{1}{1-r} \quad 2.33(1-r) = 1$$
$$1-r = 1/2.33 \quad r = 1 - 1/2.33 \quad r = 0.57$$

CLV calculation

$$CLV = \frac{M \times r}{1 + i - r} = £17.42$$

The firm should not spend £20 to attract new customers because the average profit a customer will generate for the firm is only £17.42, which would destroy value in the firm.

2. Local paper

\$1500 per year and 60 customers

\$25 per customer acquisition cost

Sales Manager

\$4500 per year ($0.1 \times 45,000$) and 50 customers

\$90 per customer

Overall

Cost of coupons : \$20 each year, each customer

Year 1

$$1500 + 4500 / 110 = \$54.54$$

+ \$20 coupon = \$74.54 acquisition cost

Years 2-4

\$20 acquisition cost

Calculation

$$\frac{M_1 - C_1}{1+i} + \dots + \frac{M_T - C_T}{1+i}$$

$$\frac{100 - 74.54}{1.07} 0.8 + \frac{150 - 20}{1.07^2} 0.75 + \frac{200 - 20}{1.07^3} 0.7 + \frac{250 - 20}{1.07^4} 0.65 = \$321.1$$